Grip or glide? The surface challenge · Staff Answers

Learning goal: I can explain how friction acts between touching surfaces and judge when grip or glide is useful.

Arrival Answers

1. Pushing a door or pulling a chair.
2. The shoe sole and the floor surface.
3. To the left, opposite the sliding.
4. Yes. You can ask or pass.

Starter & Hinge Answers

Starter: A. High friction between the brake pad and rim resists relative sliding and slows the wheel.

Hinge: A. Smooth surfaces still experience friction; whether it helps depends on the surface pair and the job.

Evidence Desk Cards

Brake pad on wheel rim: Helpful grip: high friction between pad and rim slows the wheel.

Dry metal gear teeth: Unhelpful rub: friction wastes energy as heat and wears the parts.

Treaded walking boot on wet rock: Helpful grip: friction prevents the boot slipping backwards.

Curling stone on prepared ice: Helpful glide: low friction lets the stone slide far.

Squeaky door hinge: Unhelpful rub: lubrication reduces friction so the door swings quietly.

Two mystery blocks touching: Need more information: we need the surface pair and the intended job.

Check & Repair Answer

Without friction between shoes and the ground or brake pads and wheels, we could not walk or stop safely.

Exit Ticket Answers

1. To the left (opposite to sliding). 2. Brake pad and wheel rim (or shoe sole and floor). 3. The two touching surfaces, the relative motion, and the intended job.